Aarhus University

Computer Science

Numerical Linear Algebra

Handin 10

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May 2, 2026

I'm gonna be flipping burgers in 1 year ;_;

Anyway, consider the matrix

$$A = \begin{bmatrix} 0.5 & 0.25 & 0.25 \\ 0.25 & 0.3 & 0.45 \\ 0.25 & 0.45 & 0.3 \end{bmatrix} \tag{1}$$

(a) Verify that A is symmetric and explain why A is diagonalizable with only real eigenvalues

Verifying that A is symmetric can be done by looking at it and saying 'Yep that's symmetric', since $row_i == column_i$

Or verify it using python.

```
A = np.array([[0.5, 0.25, 0.25],
              [0.25, 0.3, 0.45],
              [0.25, 0.45, 0.3]])

if np.array_equal(A, A.T):
    print(True)
# Output for reference
True
```

Per the spectral theorem 23.10 [1] any real symmetric matrix $A \in \mathbb{R}^{n \times x}$ there exist an orthogonal matrix $V \in \mathbb{R}^{n \times n}$ such that $A = V \Lambda V^T$ where Λ is a diagonal matrix, and because we can factor the matrix (1) into that, then it can be said to be diagonalizable.
To go even further, the Theorem also states that A has n real eigenvalues and an orthonormal basis consisting of it's egenvectores...Yeah take that 7-year old Gauss! I'm smart too!

(b) Verify that 1.0 is an eigenvalue of A, and then determine the other eigen-values of A.

The characteristic polynomial for A is

$$p(\lambda) = \det(A - \lambda I_3) = \det \begin{bmatrix} 0.5 - \lambda & 0.25 & 0.25 \\ 0.25 & 0.3 - \lambda & 0.45 \\ 0.25 & 0.45 & 0.3 - \lambda \end{bmatrix}$$

$$= (0.5 - \lambda) \det \begin{bmatrix} 0.3 - \lambda & 0.45 \\ 0.45 & 0.3 - \lambda \end{bmatrix} - (0.25) \det \begin{bmatrix} 0 & 0.15 - \lambda \\ 0.25 & 0.3 - \lambda \end{bmatrix} + (0.25) \det \begin{bmatrix} 0.25 & 0.3 - \lambda \\ 0.25 & 0.45 \end{bmatrix}$$
$$= ((0.3 - \lambda) \cdot (0.3 - \lambda)) - (0.45 \cdot 0.45))$$

(c) Compute an orthonormal basis for $\mathbb{R}^3$ consisting of eigenvectors of A.

(d) Determine an orthogonal matrix V and a diagonal matrix Λ such that

$$A = V\Lambda V^T \tag{2}$$

(e) For

$$v = \begin{bmatrix} 0.12 \\ 0.21 \\ 0.39 \end{bmatrix} \tag{3}$$

Use the results to determine the limits of $A^k v$ for $k \to \infty$

Appendix

References

[1] Andrew Swan. *Numerisk Linær Algebra*. Institut for Matematik, Aarhus Universitet, 2025.